Shielded Reinforcement Learning: A review of reactive methods for safe learning*

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***Abstract*— Reinforcement Learning (RL) algorithms are showing promising results in simulated environments, but their replication in real physical applications, even more so in safety-critical applications, is not yet guaranteed. Ensuring the functional safety of RL algorithms is not a trivial task since the physical integrity of the target system, also called environment, especially when there is interaction with humans, may depend on it. Among the methods recently developed with the objective of guaranteeing safety, Shielded Reinforcement Learning is defined, which defines an interaction mechanism if the action event proposed by the agent causes a non-safe state. This article provides an overview of the different Shielding Reinforcement Learning approaches. In addition to summarising the techniques used by each of them, their advantages and disadvantages are discussed. Finally, the shortcomings associated with Shielded Reinforcement Learning methods that can lead to risk or unsafe situations are discussed.**

I. INTRODUCTION

The emergence of decision-making techniques based on Machine Learning (ML) has carried out a new era in the design autonomous systems controllers. Within these techniques, one of the techniques that is recently gaining more popularity is Reinforcement Learning (RL). RL algorithms are especially linked to optimal control theory. [1, 2], as they are based on the interaction of an agent with its environment through the paradigm of trial-error and exploitation-exploration [3, 4]. Such interaction allows the agent to learn and ideally generalise the knowledge gained to deal with previously unexperienced situations [1, 5]. The agent's process of learning about the environment, i.e. mapping the *states* to the *actions* to be performed, is known as *policy*. The agent receives a *reward* for each action performed and ultimately tries to maximise the future sum of the rewards received [3, 6], thus the RL agent learning process is an policy's optimisation problem.

The great adaptability and abstraction capacity of RL methods are counterbalanced by the few or no safety guarantees they offer both in the training process and in their subsequent execution. This is why *Safe Reinforcement Learning* is defined as the process of learning policies which maximise the expected cumulative reward in problems where it is non-trivial to ensure compliance with the safety constraints in both the learning process and the implementation [7, 8, 9, 10, 11, 12, 13]. RL algorithms require exploration, which in *safety-critical* applications is often not possible or must be controlled to avoid the agent reaching risk or unsafe states. Over the past few years, many solutions were proposed to ensure safety in those RL applications intended for real-world deployment, especially in those environments shared with humans [14, 13, 15]. Many of these methods share the need to incorporate a model of the environment or deal with simple, low-dimensional environments [16].

In Embedded Critical Systems (ECS) there has been an emergence of Artificial Intelligence (AI) applications in many industrial sector such as automotive, aviation, railway, etc... Overall, it is expected AI based control systems will replace humans due to their capacity of making more accurate and faster decisions [17]. Actually, when validating an AI application the test-coverage required by current functional safety standards can not be guaranteed, not forgetting its strong dependence on training data. The requirements and methodology for functional safety industrial applications are defined in the standard IEC 61508 [18] and derivatives standards (e.g., EN 50126/8/9 for railway safety), where software and hardware safety engineering is defined. However, those standards lacks of procedures for AI based control systems. To afford this situation, among others, of overcoming this shortage VDE-AR-E 2842-61 application rule is developed, which introduces "uncertainty-related" failures for AIs [19]. However, as stated by Kläs [19] this application rule "still do not explain in detail how to deal with uncertainty-related failures when considering safety-criticality".

Therefore, it is necessary, in addition to improving technical and performance capabilities, to develop ML techniques that meet the requirements imposed by functional safety standards necessary for their implementation in industrial applications. Also, it is essential to update the state-of-the-art functional safety standards, such as IEC 61508, to the requirements that AI based controllers must comply to be considered safe their implementation on ECSs.

Among the existing methods with the objective of guaranteeing the functional safety requirements of the controllable system, there is a category called Shielded Reinforcement Learning (See Figure 1). In it, the use of a *shield* is proposed, which acts as a filter that allows only those actions that are identified as safe and if safety requirements/constraints can not be guaranteed, a safe action is provided. The safe and unsafe states are determined through constraints, so the

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decision-making problem becomes a Constrained Markov Decision Problem (CMDP).

The potential of Shielded Reinforcement Learning is that it only acts when the proposed action is not safe. In other words, the computational cost of guaranteeing functional safety requirements is minimal, making it a potential candidate for implementation in safety-critical embedded applications. The main contribution of this work includes:

- Discussion of the advantages and disadvantages of the methods reviewed.
- Categorisation of the reviewed Shielded Reinforcement Learning methods into the three Safety Levels [8].
- Identification of the shortcomings associated to the use of shields.

The rest of the paper proceeds as follow. In section 2, a brief definition of CMDP is given. In section 3, the different categories of constraints used in Reinforcement Learning problems are defined. Following, in section 4, a summary of the reviewed techniques is given. In section 5 the different techniques to implement constraints in Shielding methods are discussed. In section 6, the shortcomings showed by the Shielded Reinforcement Learning methods are discussed. Finally we summarise the main conclusions in section 7.

II. PRELIMINARIES

Within Machine Learning, the Reinforcement Learning (RL) framework is defined, which assumes that there is an agent that is situated and interacts with its environment through the paradigm of trial-error and exploration-exploitation [3]. Due to such interaction, the agent learns and generalises, and subsequently it is able to perform correctly in previously unknown situations [5] [1]. Each action taken by the agent carries with it an associated reward, which quantifies the appropriateness of both the action taken and the state it is in. Lastly, the agent learns the optimal mapping between states and actions, commonly called policy, to maximise the sum of future rewards [3]. It should be noted that RL is particularly linked to the field of control theory, especially to the optimal control theory. [1, 2].

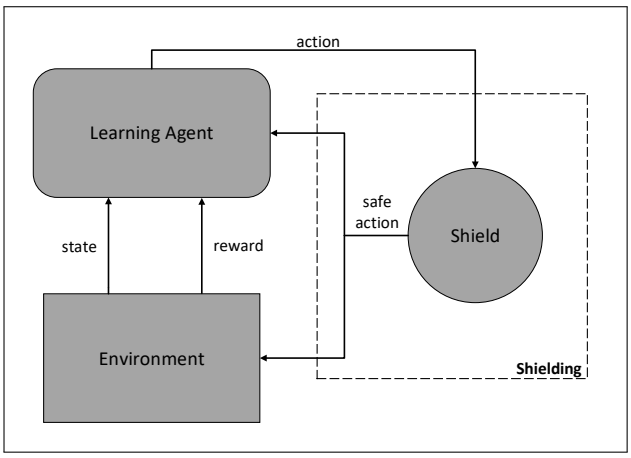

Fig. 1. Shielded Reinforcement Learning schematic

Markov Decision Processes are considered to be the idealised mathematical form of Reinforcement Learning problems.

A. Markov Decision Process

A Markov Decision Process (MDP) [4] is defined as a formalisation of sequential decision making systems, where actions determine the immediate reward, the next states and through them the future rewards. An MDP problem is described [9, 10] by the tuple $< S, A, P, R, \gamma >$, where S is the set of states, A is the set of actions, $P : S \times A \times S \rightarrow [0, 1]$ is the probabilistic transition function of the environment and $R : S \times A \times S \rightarrow \mathbb{R}$ defines the rewards function. Also, the discount factor γ determines the present value of future rewards [4].

The relationship between the states and the actions that the agent performs is given by the policy π, which is a mapping between the states and actions. The action value function $Q(s, a)$, called Q-value, is defined such that:

$$Q(s, a) = \mathbb{E}_s^{\pi}[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1})] \tag{1}$$

where $\mathbb{E}_s^{\pi}$ is the expected value from state s by applying the policy π [10]. Because the reward R quantifies at instant t the degree of success of the action executed a_t in a state s_t that transitions to state s_{t+1}, the objective of the MDP is to learn the policy π such that it maximises the cumulative discounted sum of reward:

$$\max_{\pi \in \Pi} \mathrm{J}_R^{\pi} = \max_{\pi \in \Pi} Q(s, a) \tag{2}$$

Therefore, the action chosen at each step is given by the optimisation of the reward obtained at that step and of the expect rewards to be received in the future following the policy π.

B. Constrained Markov Decision Process

Safe Reinforcement Learning is defined as the process of learning policies which maximise the expected return in problems where it is non-trivial the target system's performance and/or to ensure compliance with safety constraints in both the learning process and integration.

Such problems require safety constraints, in that case, MDP can be defined as Constrained Markov Decision Process by the tuple $< S, A, P, R, \gamma, C >$, where $c_i \in C$, $c_i : S \times A \times S \rightarrow \mathbb{R}$ are the constrained cost functions [9]. Due to the constraints, the policy space Π is reduced to those that satisfy the constraints C, obtaining the safe policy subspace Γ [20]. In such a way, only those policies that meet the imposed constraints are taken into account.

III. CONSTRAINTS

Those states and actions that pose a danger to the integrity of the environment controlled by the agent can be previously determined if the environment is known or if there is knowledge of its dynamics. In this way, the constraints for the agent can be established, so that the Reinforcement Learning

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algorithm can learn those policies that meet the requirements. Two types [9] of constraints can be defined:

- **Cumulative constraints** are those that in a given period of time, the sum or average of one of the constrained variables (e.g. constrained cost function $c_i(s_t, a_t, s_{t+1})$ [9]) cannot exceed pre-established limits.
- The **instantaneous constraints** are those constraints to be met by the chosen action at each given instant.

As far as compliance with constraints is related, three types (See Figure 2) of constraints are defined [8] which have a direct impact on the safety level of the target system:

- **Soft Constraints:** Those constraints in which the agent is encouraged to comply, but there is not any guarantee. One way of implementing these constraints is to introduce a penalty to the value-function. They are defined as Safety Level I category.
- **Probabilistic Constraints:** The maximum probability that determines the degree of compliance with the constraints by the agent is established. They make up Safety Level II.
- **Hard Constraints:** Constraints that the agent must comply with at all times. They are defined as Safety Level III.

Shielding methods are defined as those in which a *shield* is introduced, which is synthesised through the safety specifications imposed by the constraints, and through an abstraction of the dynamics of the environment [13]. Shielding is a reactive method, i.e., it only takes part in those time-steps where it foresees that the action chosen by the agent will lead to a violation of the constraints associated with safety. Thus, the premise of minimum interference is fulfilled, i.e. ensuring the safety of the target system with minimum intervention by an external entity, i.e. a shield. For this reason, the vast majority [16, 21, 12, 11, 22, 6, 13, 23, 14] of shielding methods make use of a model of the environment, and from the action proposed by the agent, they can determine the possible subsequent states in a given time horizon.

IV. SHIELDED REINFORCEMENT LEARNING

In the following lines, state-of-the-art Shielded Reinforcement Learning methods are reviewed, classifying them into the three Safety Levels taxonomy proposed by [8] and previously resumed.

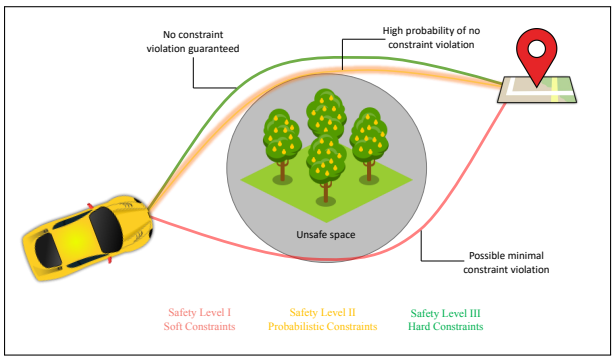

Fig. 2. Soft (Safety level I), Probabilistic (Safety level II) and Hard (Safety level III) Constraints [8].

A. SAFETY LEVEL I

*1) **Centralized and Factored Shielding**:* In environments where several agents interact, it is necessary to guarantee the safety of each agent with the environment and ensure that the agents' interaction with each other does not entail risk or unsafe situations. With this objective in mind, ElSayed-Aly et al. [11] defines the Centralised Shield, which works in a centralised manner, and the Factored Shield, which works in a decentralised procedure.

In Centralised Shield, at each instant, the shield gets the actions proposed by each agent and verifies whether they are safe. If they are safe, the action is maintained, while if not, it is replaced by another known safe action and a penalty is applied to the corresponding agent.

On the other hand, the Factored Shield works in a decentralised way. For this purpose, the state space is fragmented in such a way that each shield is in charge of a specific state subspace. Whenever an agent is at the edge of two state subspaces and decides to change to a certain one, the agent of the target state subspace checks whether the transition is safe and decides whether to allow the action.

The Centralised Shield shows limited scalability, because the increase in the number of agents means an exponential increase in the computational cost of the shield synthesis. In turn, Factored Shield sometimes offers less optimal policies than Centralised Shield. Experimental tests [11] show that the number of safety violations is drastically reduced, but not to zero, so both methods can be considered as Safety Level I.

*2) **ShieldPPO**:* In his work [24] indicates that the knowledge of the unsafe actions or constraints is not always known in advance, so sometimes they are obtained through the agent visiting the states at least once. For the shield, the author assumes that the environment is discrete, that the number of catastrophic state-action pairs is small enough to store in memory, and that the agent can observe L_ϕ simultaneously. The state-action pair labelling $L_\phi : S \times A \rightarrow \{0, 1\}$ function is defined such that $L_\phi(s, a) = 1$ if and only if taking action a in state s is safe. A non-parametric (tabular) shield is defined as follows:

$$\mathcal{S}_\mathcal{T}(s, a) = \begin{cases} 1 & (s, a) \notin \mathcal{T} \\ 0 & otherwise \end{cases} \tag{3}$$

being $\mathcal{T} = \{(s, a)\}$ a table which stores the catastrophic state-action pairs. In this way, the shield works as a mask, blocking the state-action pairs of $\mathcal{T}$. This type of shield allows the knowledge of catastrophic pairs to be shared with other agents. On the other hand, this method is unsuitable for continuous environments, where the probability of being in the same state multiple times is close to zero. The shield is integrated into the Proximal Policy Optimisation (PPO) algorithm, obtaining the variant called ShieldPPO.

In addition, in Multi Agent Reinforcemnt Learning (MARL) cases, it is interesting that the knowledge about

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the unsafe states is shared among the agents that make it up. The non-parametric shield enables to share efficiently the knowledge into all agents of the MARL problem.

Experimental results [24] shows that the number of catastrophic actions is reduced compared to the methods where shield is not applied and that over time, it has a decreasing trend due to the storage of catastrophic pairs. ShieldPPO is not a method with zero unsafe actions, so it is considered Safety Level I. As the knowledge is shared by multiple agents, the number of catastrophic pairs is reduced and the learning convergence time is drastically improved. However, it is still Safety Level I.

B. SAFETY LEVEL II

*1) **Approximate Bounded Prescience Shielding**:* One of the proposed methods [6, 16] is Bounded Prescience Shielding, where the shield accesses a simulation, such that it can determine future states at a particular finite horizon. He et al. [16] introduces a novel kind of shield called *latent shielding*. For this purpose, a Safety Recurrent State-Space Model (SRSSM) is used, which incorporates into the standard Recurrent State-Space Model (RSSM), a recurrent space-state model that makes predictions on a latent space [25], a labelling function $\mathrm{L}_{\mathcal{I}}^{\phi} : \mathcal{S}_{\mathcal{I}} \rightarrow \{safe, unsafe\}$ such that it maps latent states $s_t \in \mathcal{S}_{\mathcal{I}}$ if they correspond to a safety violation. By integrating the SRSSM they define Approximate Bounded Prescience Shielding (ABPS). ABPS differs from the standard BPS, a shielding method proposed by [6] which enumerates all traces from the current state for a bounded number of steps and labels. Each of them is labelled as safe or unsafe checking whether the probability of violation exceeds a fixed threshold ϵ. The labelling function is approximated and only a fixed number of potential trajectories of the MDP are analysed. The advantages of the ABPS are that it allows the shield to learn its own abstraction of the environment and that due to the smaller number of trajectories analysed, the prediction horizon can be increased. Even so, ABPS still suffer from the same disadvantage as the standard BPS, which is that, due to the finite prediction horizon, a trajectory that leads to a risky state beyond the prediction horizon can be categorised as safe. Therefore, these methods are categorised as Safety Level II.

*2) **Probabilistic Shield**:* As the name implies, Probabilistic Shield [12] uses probabilistic constraints.The action is blocked when it increases the probability of violating a safety constraint beyond a threshold δ. They pose the problem environment as an *arena* where one controllable agent (the avatar) and several uncontrollable agents (the adversaries) operate. The constraints are defined using Linear Temporal Logic (LTL) properties, so that the model uses value iteration or linear programming to compute probabilities and determine compliance with LTL properties. A *behaviour model* is constructed for each adversary, observing their behaviour in smaller scenarios. With the models and a concrete arena, they build an MDP that they call *safety-relevant MDP quotient*. The shield construction is carried out using the action values, to form the *δ-shield*, which determines the optimal actions for the given safety specifications and is stored in a look-up table. The shield blocks unsafe actions. This method offers a Safety Level II, so that non-compliance with safety constraints can be limited. On the negative side, it should be noted that the safety-relevant MDP quotient is built for a trained scenario. So, safety may not be guaranteed in new scenarios (previously unseen ones).

C. SAFETY LEVEL III

*1) **Base Shielded RL**:* The use of shields was proposed by Alshiek et. al. [13] for the first time. The idea was based on leaving an agent in execution on its own, and only interfering in case of violation of the safety constraints. Actions that does not comply with the constraints are blocked and a safe action are offered instead. It is defined as a reactive system, so it uses a finite Mealy machine and the safety specifications, or safety constraints, are defined using Linear Temporal Logic (LTL). To determine whether an action leads to a violation of the constraints, it makes use of a environment model. To synthesise the shield, the use of a *safety-game* is proposed so that the two opponents are the environment model, which chooses the next observation from the MDP, and the agent, which chooses the action. A *play* is won by the agent if only safe states are visited during the play [13]. The winning region, the state set from which a winning strategy exists, is computed and the shield is synthesised. Despite incorporating the shield, the Shielded MDP's convergence is guaranteed because it is still a standard MDP. The experiments developed by Alshiek et al. [13] compares a shield with and without negative reward feedback. The results shown that the inclusion of the negative reward feedback induces slower learning rate and the agent to make unsafe actions. This way, the shield with negative reward feedback is categorised as Safety Level I and the shield without one is categorised as Safety Level III.

*2) **Model Predictive Shielding**:* For deterministic and nonlinear environments, Bastani [22] proposes the use of a method called Model Predictive Shielding (MPS). This method determines for each step whether the next state s_{t+1} obtained after applying the action a_t is recoverable employing the *backup policy*. If the assumption is met, it allows the action offered by the agent to be applied. In case the recoverability is not guaranteed, the backup policy is applied to bring the robot to an equilibrium point. The backup policy is made up of two policies:

- **Recovery Policy:** The control application is moved from the state where it is located to the nearest equilibrium point using a predefined π_{rec} policy. Although sometimes there is an obvious choice (e.g., in a robotic arm it would be to stop it completely), it may happen that due to the great complexity of the environment, these equilibrium points are not so easy to determine. It is proposed the use of RL to train this policy π_{rec}.
- **Equilibrium Policy:** Once the control application has been brought to the equilibrium point, it must be ensured that it stays there using a predefined π_{eq} policy.

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Several actions may guarantee to keep the robot at equilibrium, so one of them is selected randomly.

In the case of unstable equilibrium points, feedback stabilisation is used to ensure the control application's safety. The policy π_{eq} is used to be the Linear Quadratic Regulator (LQR) as a linear approximation of the dynamics around the unstable equilibrium point, so that later is computed by LQR verification the states for which π_{eq} is safe.

The experiment conducted in the cart-pole environment [22] provides satisfactory results, since it guarantees total safety both in the original and modified tests. For this reason, MPS is classified as Safety Level III. However, it should be noted that although the choice of a backup policy ensures compliance with safety requirements, it may not be an optimal policy since the backup policy is defined by the developer which tends to be conservative. Moreover, in the case of the modified cart-pole test, it is observed that the reward obtained is not as good.

3) ***Failsafe Planner****:* In collaborative robotics environments, due to the inherent stochasticity of humans, it is complex to guarantee safety, even in cases where safety constraints are correctly defined. In order to guarantee the safety shield concept of *speed and separation monitoring* (SSM) defined in DIN EN ISO 10218-1 2021, 5.10.3, which states that the robot has to stop completely in case of a possible collision with a human, [14] proposes the use of a safety shield which combines the low-frequency RL agent with formal high-frequency safety verification algorithm.

Author proposes the use of a RL Soft Actor-Critic (SAC) agent combined with Hindsight Experience Replay. On one hand, the agent's actions determine the sets of *intermediate goal* joint positions to reach the final goal. The agent actions are executed for user-defined Δt time or until all joint positions are within ϵ-range of an intermediate goal.

On the other hand, the shield is composed of the long-term planner and the high-frequency safe trajectory *Failsafe Planner*. The long-term planner calculates the *intended trajectory* from the current joint state to the intermediate goal. In turn, the Failsafe Planner, while executing the trajectory between t_k and t_{k+1}, computes a *failsafe trajectory* starting at t_{k+2}. If the latter path cannot be guaranteed as failsafe trajectory, the subsequent path is executed. To compute the failsafe trajectories, the occupancy space of both the robot and the human is modelled using capsule form spaces to speed up the computation time. It is defined the space that the human can occupy between instants $[t_k, t_{k+n}]$ by reachable occupancy $\Gamma([t_k, t_{k+n}])$, where n is the number of time instants required for the robot to stop. If it is observed that no capsule corresponding to the robot enters $\Gamma([t_k, t_{k+n}])$ in that time horizon, the trajectory is considered failsafe.

Authors [14] state that in the experiments the robot with the Failsafe Planner fully complies with the safety requirements. Therefore, Failsafe Planner is categorised as Safety Level III. Even so, the modelled human movements are quite limited, so safety is not guaranteed in more complex environments. In addition, the computational cost required to determine the position in space of the capsules and their collisions, which is expected to be non-trivial, is not specified.

4) ***POMDP Shielding****:* As a first shielding method developed for those Partially Observable Markov Decision Processes (POMDP) problems, [23] presents a shield integration computed via satisfiability solving with Tensorflow's TF-Agents RL algorithm package. A POMDP is defined by the tuple $\mathcal{M} = \langle S, A, O, Z, P, R \rangle$ where Z is a finite observation space and $O(z|s)$ is the probability of observing z when the environment is at state s [23]. The policy is defined as $\pi : (Z \times A)^* \times Z \rightarrow Dist(A)$ and the agent may depend on the past actions taken $(Z \times A)^*$.

Safety constraints are defined via *reach-avoid* $\varphi = \langle REACH, AVOID \rangle$ specifications. Because in a POMDP the agent must estimate the value of the state using observations, it is defined a *belief* $\mathfrak{b} : (Z \times A)^* \times Z \rightarrow Dist(S)$, where Z is the space of observations, which is a distribution describing the probability that the agent is in a certain state based on the history. The set of the *winning beliefs*, those beliefs that satisfy $\Pr^{\pi}_{\mathfrak{b}}(AVOID) = 0$ and $\Pr^{\pi}_{\mathfrak{b}}(REACH) = 1$, are defined as the winning regions or also called safe region used to synthesise the shield.

Experimental results [23] shows that shields guarantee full compliance with safety specifications, so it can be classified as a Safety Level III method. In addition, it is observed that shielding accelerates convergence because it limits the state space, but also increases the computational cost. In addition, Carr et al. concludes that shields do not enforce reaching targets quickly and can degrade the performance of the agent. Finally, it should be noted that results show that a shield is more effective in some RL algorithms (PPO, SAC) than in others (DQN and DDQN).

V. SHIELDING METHODS DISCUSSION

In the following Table 1 the algorithms previously reviewed are classified according to the Safety Level, the learning environment attributes (Characteristics, simulation or real, dimensionality and deterministic) and whether shield gives feedback as negative reward (punishment) to the agent, as shown in Table I.

Four Safety Integrity Levels (SILs) are defined within the functional safety standard IEC 61508, which quantify the probability of failure per hour (On high demand or continuous mode of operation) in the execution of a given safety function. SIL 1 is the lowest Integrity Safety Level, while SIL 4 is the highest [26]. Therefore, the guarantee that the target system does not enter unsafe states is bounded according to the IEC 61508 standard, so only methods corresponding to Safety level II and III can guarantee the different SIL levels. However, the methods listed in Safety Levels II and III have only proven their suitability in simulated environments and are therefore not yet guaranteed to be suitable for implementation in safety-critical industrial applications.

One of the negative aspects of the methods reviewed is that the experimentation to obtain the results has been carried out in low-dimensional, deterministic environments that do not fit to real industrial applications conditions or requirements.

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TABLE I

SHIELDED REINFORCEMENT LEARNING METHODS CLASSIFICATION

Shielding method	Safety Level	ENVIRONMENT Characteristics	Sim/Real	Dimensionality	Deterministic	Feedback
Base Shielded RL	Safety Level I/III	GridWorld, Self-Driving car Scenario and Water tank scenario	Simulation	Low, High and Low	-	✓/✗
Centralised and Factored Shielding	Safety Level I	GridWorld and Cooperative Navigation	Simulation	Low	✓	✓
ShieldPPO	Safety Level I	LavaGrid	Simulation	Low	✓	✗
ABPS	Safety Level II	Visual Grid World and Cliff Driver	Simulation	High and Low	✓	✓
Probabilistic Shield	Safety Level II	PAC-MAN arcade game and Warehouse Floor Plan	Simulation	Low	✓	✗
Model Predictive Shielding	Safety Level III	Cart-Pole	Simulation	Low	✓	✗
Failsafe Planner	Safety Level III	Custom robot environment built in Open Dynamics Engine	Simulation	High	-	✗
POMDP Shielding	Safety Level III	GridWorld	Simulation	Low	✓	✓

Although benchmarks based on Atari games, Gridworlds or OpenAI Gym are adequate to compare the performance of the developed algorithms, they still show a large gap with real industrial applications characteristics (e.g. dimension, stochasticity, noise, etc.). This is the reason behind the need to establish high level complexity benchmarks that can offer a closer approximation to safety-critical industrial applications. The counterpart is that the computational cost of learning process of Shielded reinforcement learning algorithms will grow exponentially. For this purpose, there are libraries (e.g. Nvidia Isaac Gym) that make use of parallel computing. These environments allow thousands of systems to be simulated simultaneously, offering an API based on the vectorisation of observations and actions. In the case of Isaac Gym, due to its compatibility with Nvidia Omniverse platform, it offers the possibility to generate high-capacity and scalable environments, close to safety-critical industrial applications.

Only Base Shielded RL, Centralised and Factored Shielding, ABPS and POMDP Shielding methods shows some feedback to the agent concerning the proposed action. If the action proposed by the agent turns out to be rejected by the shield, the agent receives a penalty, a negative reward, so that in the future it will avoid proposing the same action again [16]. This approach is interesting, because if the agent learns from the feedback of the shield the actions to avoid (the agent becomes biased), the intervention of the shield will be lower in the future [23], which means a lower computational cost.

Regarding multi-agent problems (MARL), Centralised and Factored Shielding focuses mainly on the interaction between agents and how to prevent their interaction from producing risky states. On the other hand, ShieldPPO also deals with MARL problems, but from the point of view that sharing what an agent learns with other agents is beneficial for the whole ecosystem. These approaches, despite their limitations such as high computational cost, are of great interest because many industrial applications involve the use of a large number of similar machines working in a shared environment. However, all shielded MARL methods are categorised as Safety Level I, which are insufficient for the industrial control application standards. Research of new shielded MARL methods that can be categorised as Safety Level II or III seems to be necessary.

It should be noted that of all the methods reviewed, Failsafe Planner is the only one that takes into consideration the functional safety standards defined for Machine Learning-based methods applicable to industrial processes. Specifically, it takes into account the DIN EN ISO 10218-1 2021, 2.10.3, linked to the functional safety standard for robots and robotic devices.

As it has been observed, the vast majority (ABPS, Centralised/Factored Shield, Probabilistic Shield, MPS, Failsafe Planner, POMDP Shielding) of the all shielding methods reviewed make use of models that contain information about the dynamics of the environment. This implies that, in order to correctly guarantee the safety of the target system, the model must be as similar as in the real system. This is not always possible, because the agent may encounter previously unseen environments. On the other hand, ShieldPPO method does not make use of any model associated with the shield, and the learning of the insecure actions is done by visiting each of the insecure states once. If the learning is done in simulated environments or, in the worst case, in real ones that do not entail serious risks to the integrity of the target

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system, it can be a valid method. However, it is extremely important in this method that the agent visits all the risk states before deployment in the real world application, as failure to identify one of these states in the learning process can lead to experiencing it in the execution process.

In the agent's learning process, the model associated with the shield is also constructed or refined. This means that in the learning process, the shield can incorrectly identify the actions that lead to a safe or unsafe state. To this end, [16] proposes training the agent without the shield and gradually introducing it. This minimises the possibility of the shield model incorrectly identifying a safe or unsafe state in early learning states. The disadvantages of this are that i) Training must be done in a simulation environment or, in the worst case, in a real low-risk environment, and ii) The improvement in convergence shown by the use of the shield is lost during the learning process.

Last but not least, the taxonomy concerning to the Safety Level given in this paper is only applicable for environments without changes on its dynamics. In case of changes in environment dynamics, it is not guaranteed that each method will keep the same indicated Safety Level.

VI. SHORTCOMINGS OF SHIELDING

It has been shown that Shielded Reinforcement Learning methods can adapt to changes in the dynamics of the environment because they incorporate a model that is updated by the observation of the execution environment. This partially guarantees the robustness necessary in safety-critical applications, but there is still the problem of constraints and the speed of model adaptation. The same problem arises in the transition of the RL agent from the simulation to the real environment due to the simulation to real gap [27]. So, there is not guaranteed that an agent that complies with the safety constraints in simulation, will continue to comply with them once the algorithm is implemented in the cyber-physical system to be controlled.

In the case of RL, [10] identifies 3 zones in terms of safety: safe, hidden unsafe and unsafe states. Safe and unsafe states are, as their names indicate, those states where the environment is considered safe or unsafe with respect to the safety constraints imposed. But there is a third zone, the hidden unsafe states, which corresponds to those states where even if the imposed safety constraints are met, any action $a \in A$ leads to an unsafe state or another hidden unsafe state. It should be noted that the hidden unsafe states are points of no return, because sooner or later they lead to the agent ending up in an unsafe state.

Let it be a system controller that is able to identify all safe, unsafe and hidden unsafe actions through the inherent model of the shield. In other words, we have an algorithm classified as Safety Level III. An action a_t proposed by the agent in state s_t causes the environment to transition to state s_{t+1} (See figure 3), i.e. a safe state. Therefore, when acting correctly, the shield does not block the action offered by the agent. Let's suppose now that, due to an unusual event, the dynamics of the environment is modified drastically. This

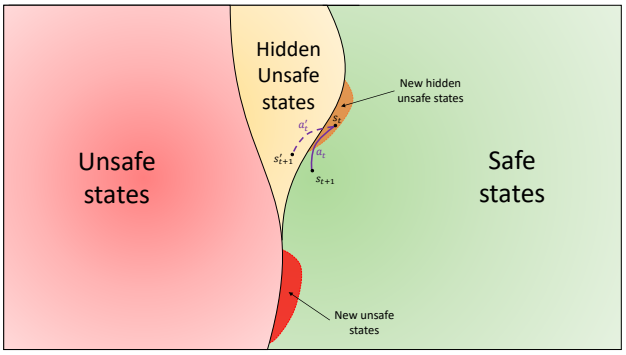

Fig. 3. The same action a_t applied in state s_t may imply that the environment evolves to the safe state s_{t+1} (Initial environment) or to the hidden unsafe state s'_{t+1} (Drastically modified environment) depending on the dynamics of the environment.

means that from the same state s_t, and applying the same action a_t proposed by the agent, the environment transitions to the state s'_{t+1} which is a unsafe or a hidden unsafe state. As the model associated with the shield corresponds to the instants before the unusual event, the shield will continue to identify the action as safe, which leads to the environment ending up in a catastrophic state. This unusual event implies that the space of safe, unsafe and hidden unsafe states is modified. Several states that previously corresponded to safe states now correspond to unsafe states (New unsafe states) or hidden unsafe states (New hidden unsafe states). As a result, the model associated with the shield is outdated, so that there is a vulnerability in terms of blocking unsafe actions. This means that the algorithm used is now Safety Level I.

Over time, both the agent and the shield will adapt to this new dynamic environment, but the time needed to adapt is not trivial. Moreover, a main question arises from a functional safety point of view: during the period of adaptation to the new environment, i.e. the relearning period, to what extent are guaranteed the safety constraints that previously were guaranteed. Although there are studies [21] that aim to measure this problem by employing shield adaptability, they have not yet been developed for Safe Reinforcement Learning problems. This is why a new research line is proposed where a methodology to mitigate the vulnerability of these transient periods of Shielded Reinforcement Learning algorithms is proposed.

VII. CONCLUSIONS

Shielded Safe Reinforcement Learning methods can guarantee partial or complete agent safety, but do not yet guarantee the necessary robustness for their application in cyber-physical systems. Even so, they show potential to be integrated into safety-critical embedded systems since they are reactive systems that reduce the associated computational cost. Despite the promising results, the existing literature has simple or low-dimensional problems as evidence, so studying the feasibility of stochastic problems with rapid changes in the environment and high dimensions is necessary. On the other hand, it is necessary to propose methodologies that

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allow the verification and validation of these algorithms, if they are going to be integrated into real-world (critical) control systems. Also, in functional safety standards, such as IEC 61508, the inclusion of functional safety requirements that AI based controllers must fulfil is crucial. Finally, in this paper, in order to improve the robustness of the control systems, a research line is proposed which aims to minimise the safety vulnerability associated to the drastic or unexpected changes in the dynamics of the environment, or, to the transition from simulated to the target environment.

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